# Does the black hole get bigger?

Yes. Mass that falls in change the geometry of the black hole;


Q: given a distribution of mass, how do I generate a metric?

## Cosmology

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + \Lambda g_{\mu\nu} = 8\pi G_N T_{\mu\nu}$$

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = 8\pi G_N \left( T_{\mu\nu} - \boxed{\frac{\Lambda g_{\mu\nu}}{8\pi G_N}} \right).$$

vacuum energy

① What's $T_{\mu\nu}$?

② What are the symmetries?

$$T^{\mu\nu} = \begin{pmatrix} \rho & & & \\ & p & & \\ & & p & \\ & & & p \end{pmatrix} \overset{S.R.}{=} (\rho + p) U^\mu U^\nu + p\, \eta^{\mu\nu}$$

perfect fluid

four-velocity

verify: at rest, $U^\mu = \begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$

In GR, $T^{\mu\nu} = (\rho + p) U^\mu U^\nu + p g^{\mu\nu}$

3 types of "fluid"

(a) $p = 0$, "dust" (pressure-less matter)

(b) $p = \frac{1}{3}\rho$, "radiation" (magnetic waves carry momentum in addition to energy)

$$T^{\mu\nu} = \left(F^{\mu\tau} F_{\tau}{}^{\rho} - \frac{1}{4}\eta^{\mu\nu} F^{\sigma\tau} F_{\sigma\tau}\right)$$

constant ?

what magnetism ?

$T^{\mu}{}_{\mu} = 0$ (traceless)

$$T^{\mu}{}_{\mu} = 0 = -\rho + 3p \Rightarrow p = \frac{1}{3}\rho$$

(c) $p = -\rho$ "vacuum energy"

comes from the RHS.

why are we setting them equal?

$$\boxed{\frac{\Lambda}{8\pi G_N}\begin{pmatrix} -1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix} = \begin{pmatrix} \rho & & & \\ & p & & \\ & & p & \\ & & & p \end{pmatrix}}$$

$\hookrightarrow$ $\rho = -\frac{\Lambda}{8\pi G_N}$

$p = \frac{\Lambda}{8\pi G_N}$

$\rho = -p.$

(2) Symmetries

"Copernican Principle" the Universe is uniform.

(a) homogeneity: for every point p there are symmetries that take p to a near-by point.

for d-dim: d translations

(b) isotropy: for any point p, maximal rotational symmetries about p: $\frac{d(d-1)}{2}$. symmetries

$$\binom{d}{2} = \frac{d!}{(d-2)!\,2!} = \frac{d(d-1)}{2}.$$

↳ how does it relate to psets?

(a)+(b): $d + \frac{d(d-1)}{2} = \frac{d(d+1)}{2}$ symmetries

"maximally symmetric"

↳ only 3 such spaces

(1) Flat $R=0$

(2) de Sitter space $R>0$

(3) anti-de Sitter $R<0$.

$$g^{\alpha\gamma} g_{\alpha\delta}\, g_{\beta\gamma}$$
$$= g^{\gamma}{}_{\delta}\, g_{\beta\gamma} = \underline{g_{\beta\delta}}$$

③

$g^{\alpha\gamma} g_{\alpha\gamma} = d.$

derive this.

max symmetric space:

$$R_{\alpha\beta\gamma\delta} = \frac{R}{d(d-1)} \left( g_{\alpha\gamma} g_{\beta\delta} - g_{\alpha\delta} g_{\beta\gamma} \right)$$

$$R_{\beta\delta} = g^{\alpha\gamma} R_{\alpha\beta\gamma\delta} = \boxed{\frac{R}{d(d-1)} (d-1) g_{\beta\delta}} = \frac{R}{d} g_{\beta\delta}$$

$$R = g^{\beta\delta} R_{\beta\delta} = R.$$

we can only accomodate $T_{\mu\nu}$ ~~accomodate~~ that's $\propto$ to $g_{\mu\nu}$.

$$R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R = R \left( \frac{1}{d} - \frac{1}{2} \right) g_{\mu\nu}$$

Hence assumptions (a) & (b) are too restrictive.

## New symmetries

$\boxed{ds_3^2}$

$$ds^2 = -dt^2 + R^2(t) \, g_{ij} \, dx^i dx^j$$

3-d max symmetric.

$$ds_{(3)}^2 = \rho^{2\beta(r)} dr^2 + r^2 \left( d\theta^2 + \sin^2\theta \, d\varphi^2 \right)$$

there's mathematica packages.

$$R^{(3)}_{\bar{r}\bar{r}} = \frac{2}{\bar{r}}\frac{d\beta}{d\bar{r}}$$

constant

$$R_{\bar{r}\bar{r}} = \frac{2}{\bar{r}}\frac{d\beta}{d\bar{r}} = 2ke^{2\beta} \quad (R = cg)$$

constant

$$e^{-2\beta} d\beta = k\bar{r}\, d\bar{r}$$

$$-\frac{1}{2}d(e^{-2\beta}) = d\left(\frac{1}{2}k\bar{r}^2\right)$$

$$\Downarrow$$

$$\boxed{e^{-2\beta} = C - k\bar{r}^2}$$

by choice of coordinates, set $C=1$.

$$e^{-2\beta} = 1 - k\bar{r}^2$$

$$ds^2 = -dt^2 + R(t)^2\left(\frac{d\bar{r}^2}{1-k\bar{r}^2} + \bar{r}^2 d\Omega^2\right)$$

$$ds^2_{(3)} = e^{2\beta(r)} d\bar{r}^2 + \bar{r}^2\left(d\theta^2 + \sin^2\theta\, d\varphi^2\right)$$

new coordinates

Robertson-Walker

$$\boxed{ds^2 = -dt^2 + a^2(t)\left(\frac{dr^2}{1-Kr^2} + r^2 d\Omega^2\right)}$$

$K=0$ flat
$K=+1$ closed
$K=-1$ open

$a(t)$ defined such that $a(t=\text{now}) = 1$.

scale factor

want to make sure that $\nabla_\mu T^\mu{}_\nu = 0$

$$T^{\mu\nu} = (\rho + p) u^\mu u^\nu + p g^{\mu\nu}$$

why ?

$$T^\mu{}_\nu = (\rho+p) u^\mu u_\nu + p \boxed{\delta^\mu_\nu}$$

$$\boxed{g_{\nu\nu} g^{\mu\nu} = g^\mu_\nu \ ?}$$

$$= \begin{pmatrix} -\rho & & & \\ & p & & \\ & & p & \\ & & & p \end{pmatrix}$$

$$\boxed{\begin{aligned} \nabla_\mu T^a &= \partial_\mu T^a \\ &+ \Gamma^?_{??} T^a \end{aligned}}$$

$\Downarrow$

I need exercise on how to take covariant derivatives

$$\underbrace{\partial_\mu T^\mu{}_0 + \Gamma^\mu_{\mu\tau} T^\tau{}_0 - \Gamma^\tau_{\mu 0} T^\mu{}_\tau}$$

omg derive this:

Can you calculate the Christoffel symbols yourself?

$$\Gamma^1_{10} = \Gamma^2_{20} = \Gamma^3_{30} = \frac{\dot a}{a}.$$

$\frac{da}{dt}$.

$$= \frac{d\rho}{dt} + 3\frac{\dot a}{a}(-\rho) - 3p\frac{\dot a}{a}$$

energy conservation:

$$\frac{d\rho}{dt} = -3(\rho+p)\frac{\dot a}{a}.$$

if we express $p$ in terms of $\rho$.

$$p = \omega\rho.$$

$$\frac{d\rho}{dt} = -3\rho(1+\omega)\frac{\dot a}{a}$$

$$\frac{\dot\rho}{\rho} = -3(1+\omega)\left(\frac{\dot a}{a}\right)$$

$\ln\rho \sim -3(1+\omega)\ln a$

$\hookrightarrow \rho = \rho_0 a^{-3(1+\omega)}$

$p = 0 \Rightarrow \omega = 0 \Rightarrow \rho = \rho_0 a^{-3}$

$p = \frac{\rho}{3} \Rightarrow \omega = \frac{1}{3} \Rightarrow \rho = \rho_0 a^{-4}$

$p = -\rho \Rightarrow \omega = -1 \Rightarrow \rho = \text{constant}.$

Friedmann's Eqns.